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(54) METHOD OF BACKFLOW PREVENTION IN A GAS DELIVERY LINE

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(63) Continuation of application No. 18/416,609, filed on Jan. 18, 2024, which is a continuation-in-part of application No. 17/453,731, filed on Nov. 5, 2021, now abandoned.

(60) Provisional application No. 63/110,522, filed on Nov. 6, 2020.

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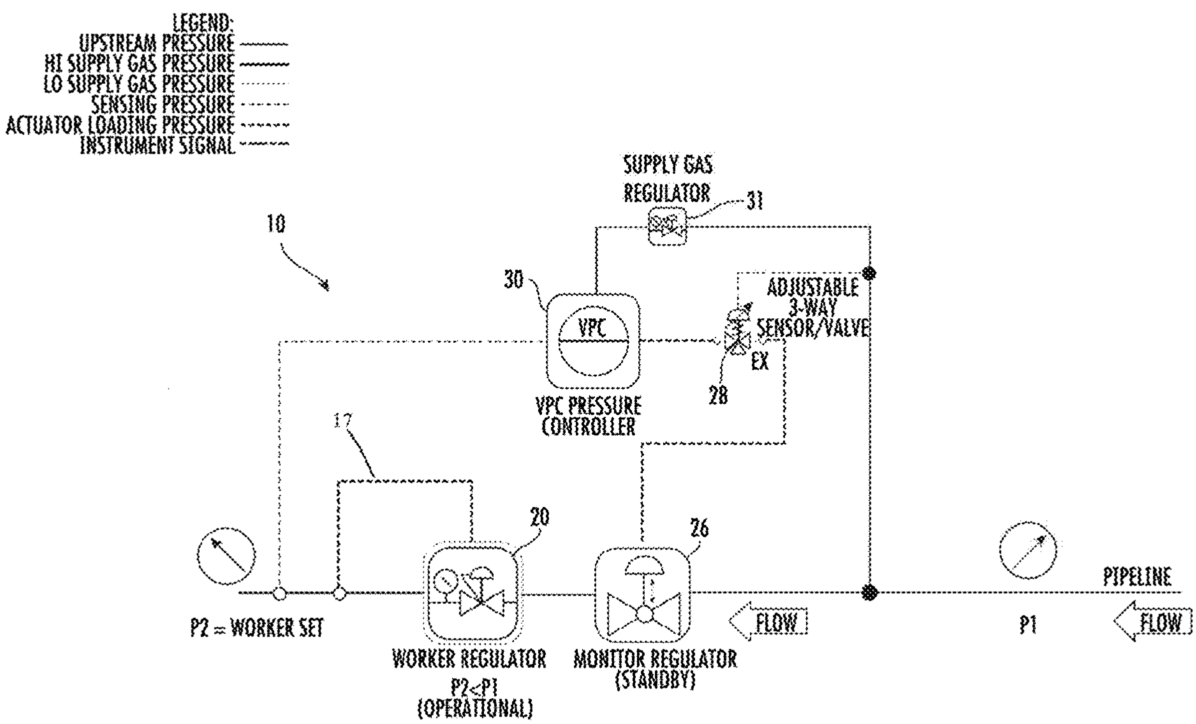
(52) **U.S. Cl.**
CPC ***G05D 7/0629*** (2013.01); ***G05D 16/024*** (2019.01); ***G05D 16/028*** (2019.01)

(57) **ABSTRACT**

A method for preventing flow reversal due to an upstream pressure drop in a natural gas supply line. The method, using a unique system configuration, includes setting a threshold low pressure for the upstream gas flow, continually sensing the upstream pressure, activating a trigger valve when the upstream pressure falls below the threshold low pressure, and closing a control valve in response to the trigger valve to prevent reversal of flow in the gas supply line.



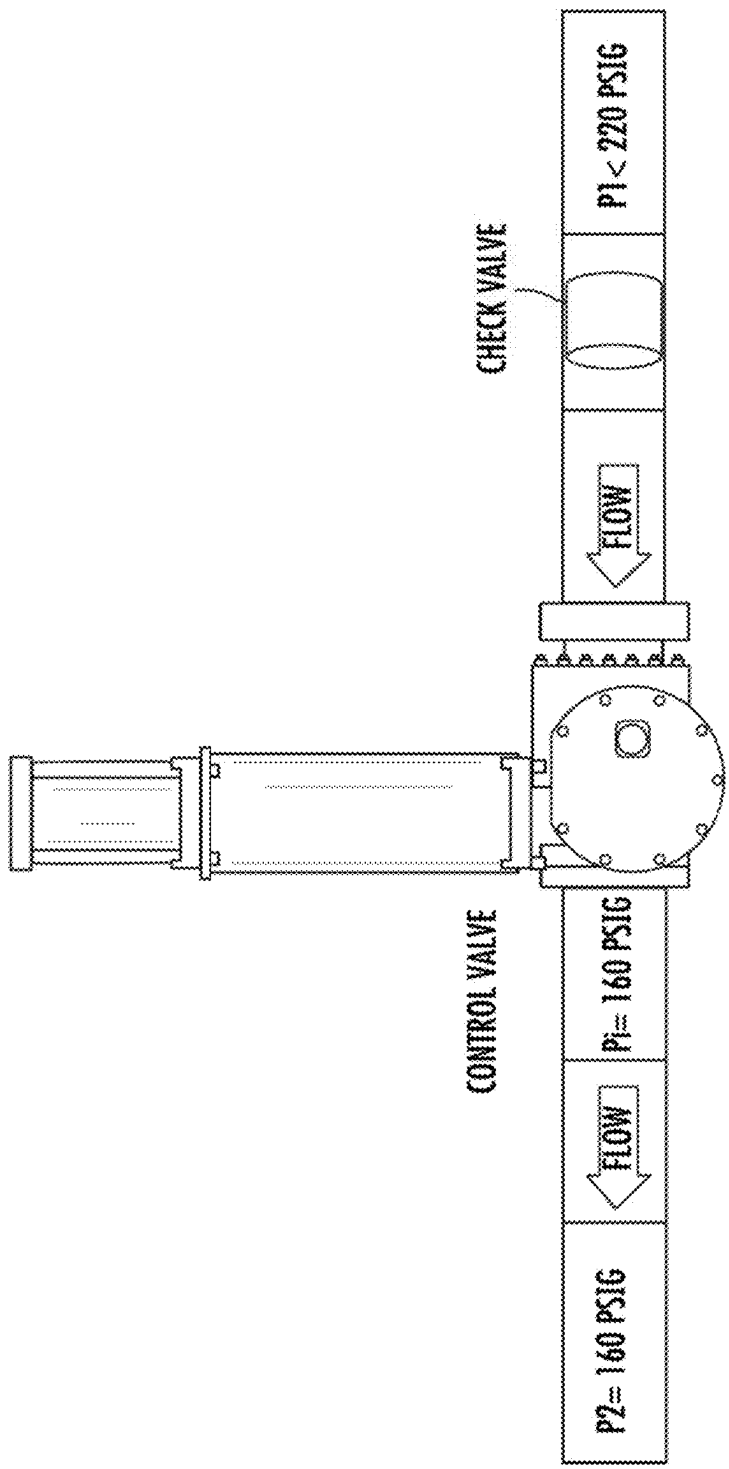

P1 < 220 PSIG
CHECK VALVE
FLOW
CONTROL VALVE
Pi= 160 PSIG
FLOW
P2= 160 PSIG

FIG. 1

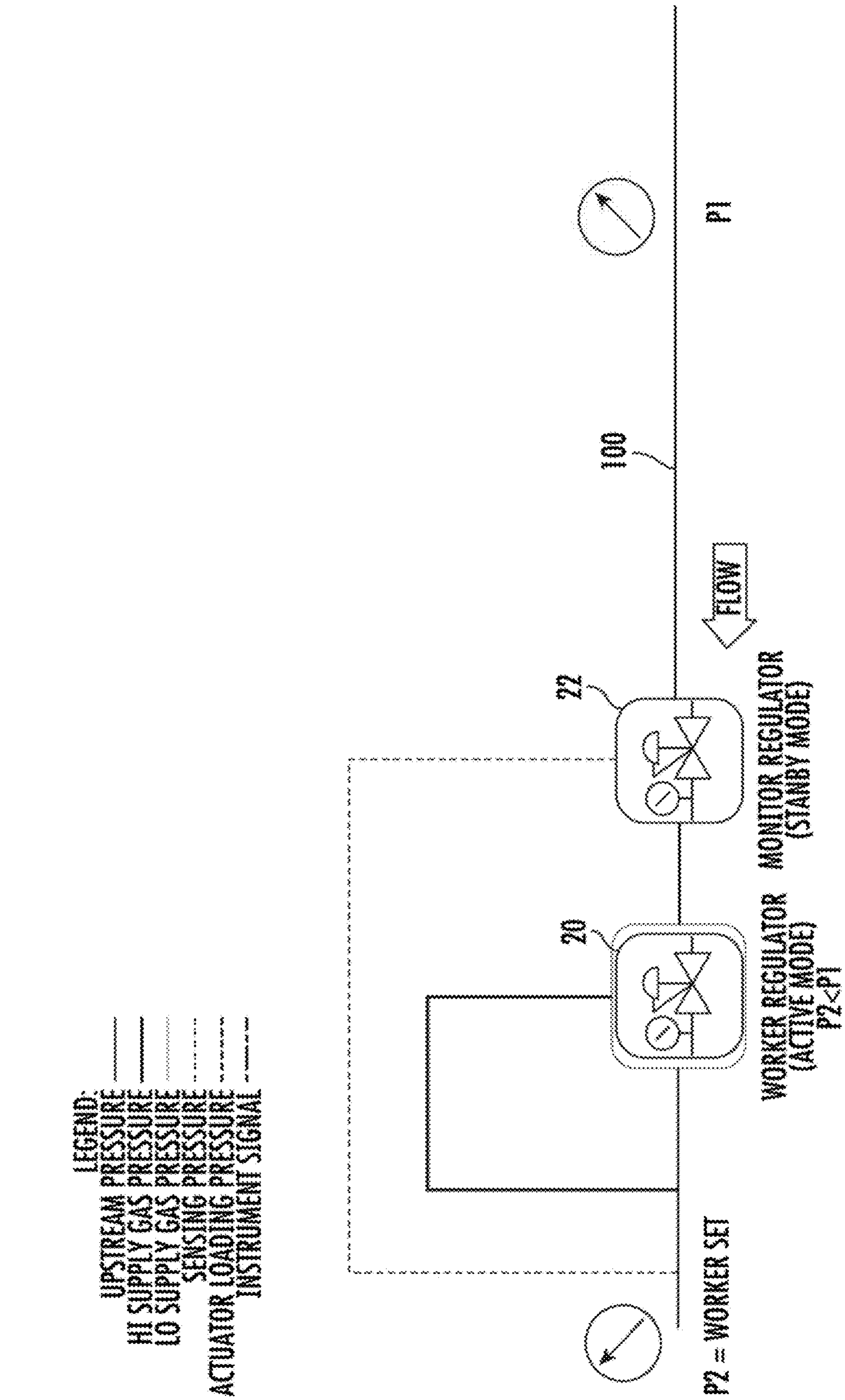


FIG. 2
PRIOR ART

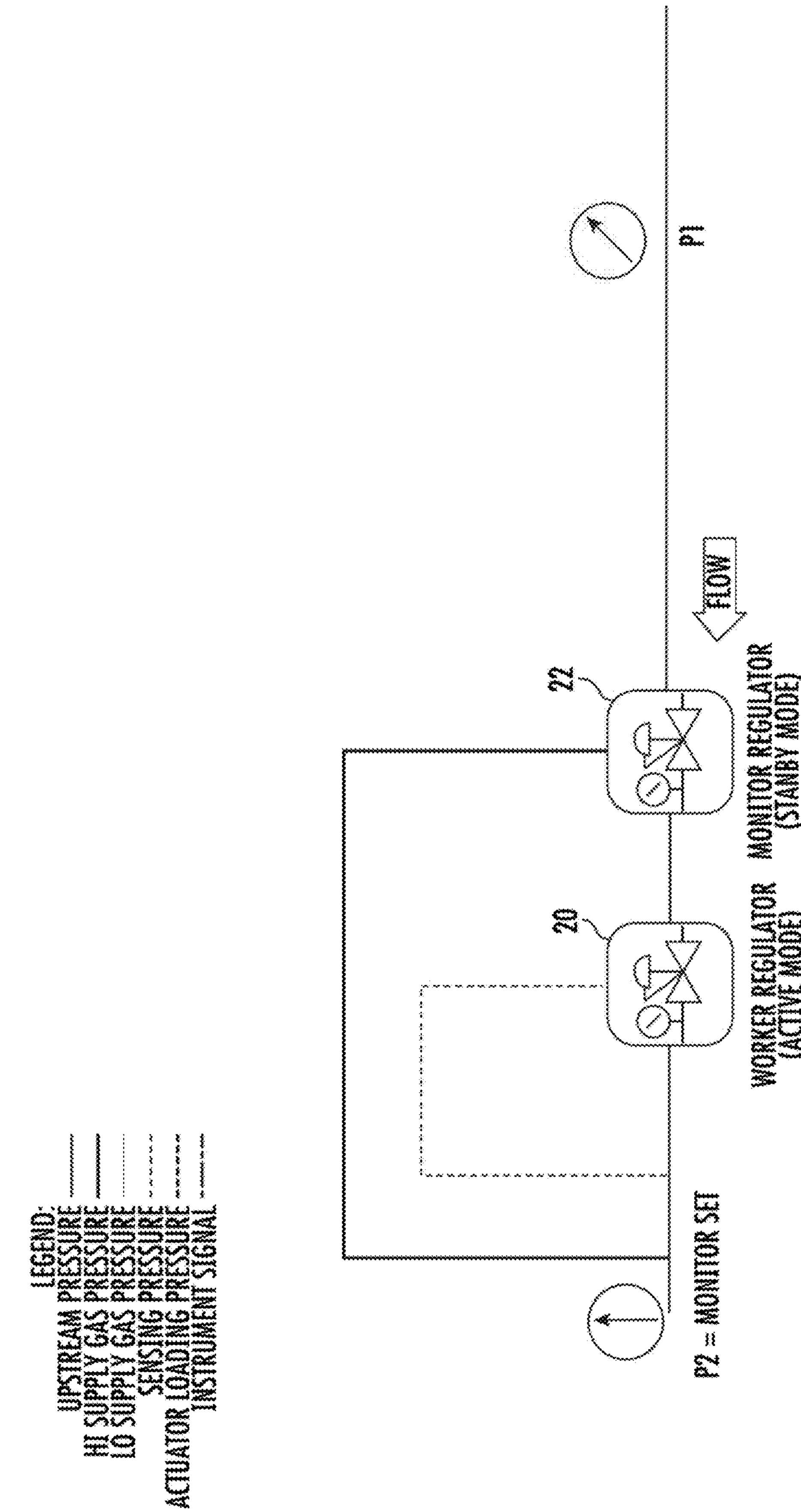


FIG. 3
PRIOR ART

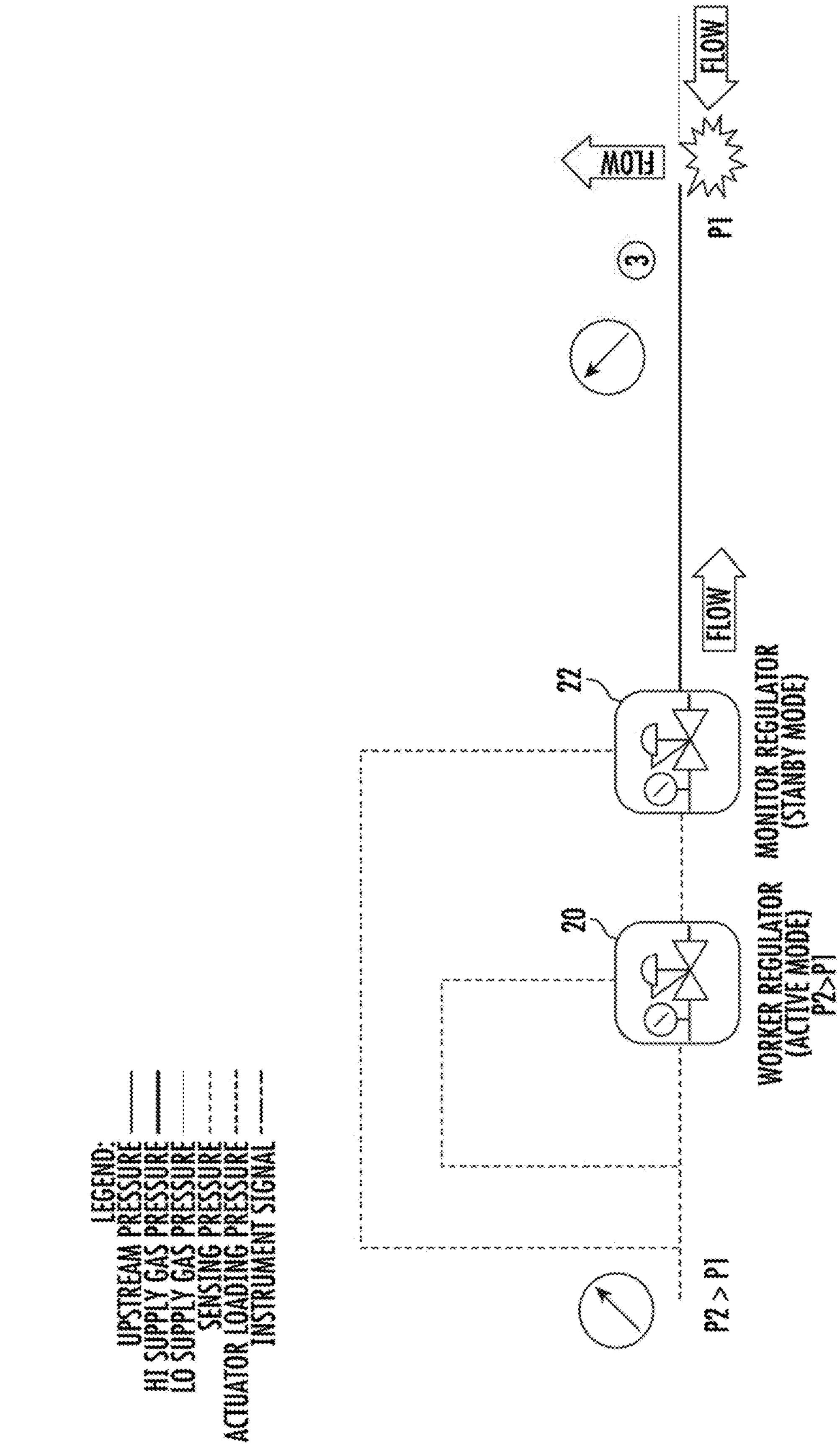


FIG. 4
PRIOR ART

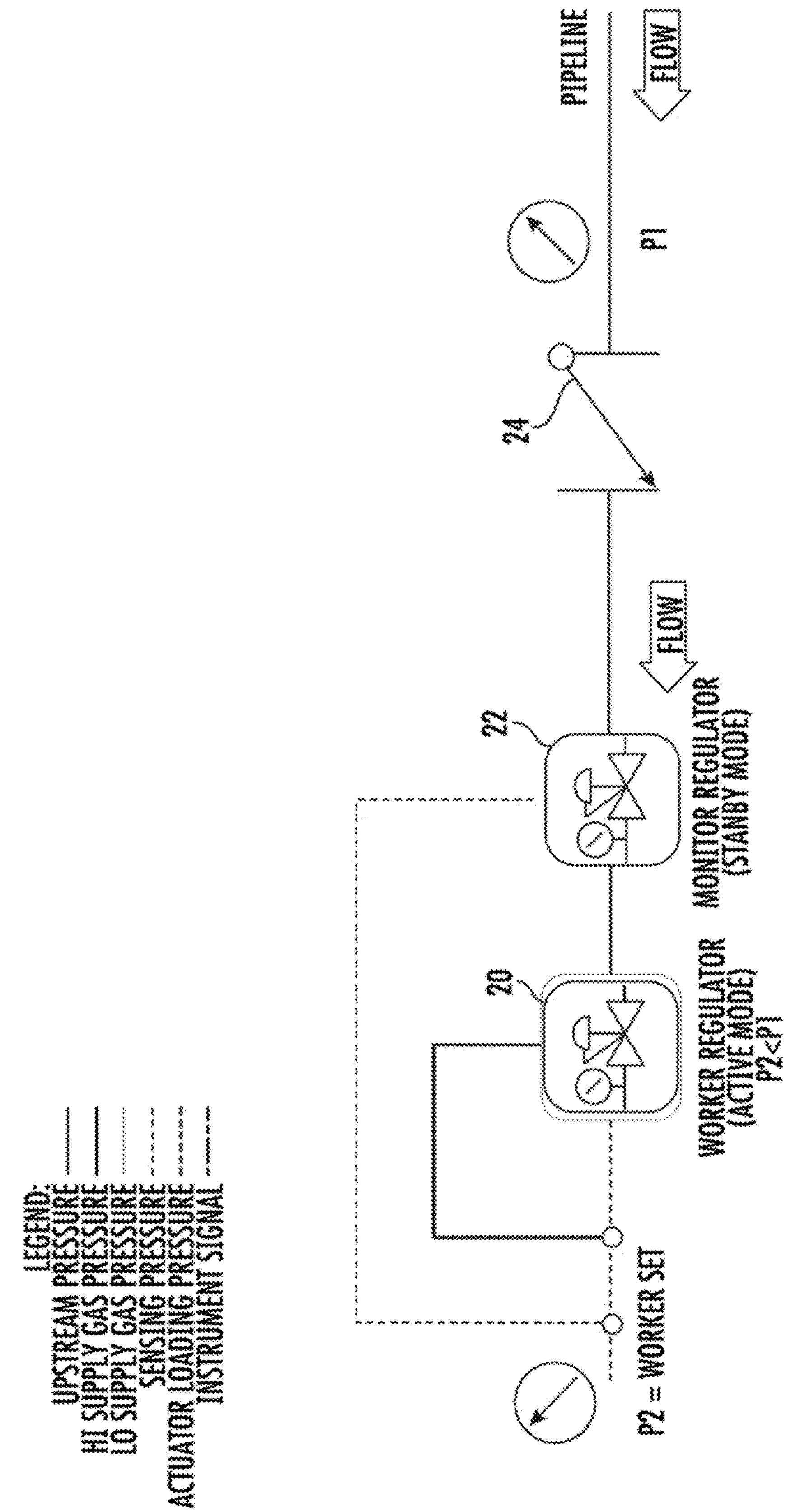


FIG. 5
PRIOR ART

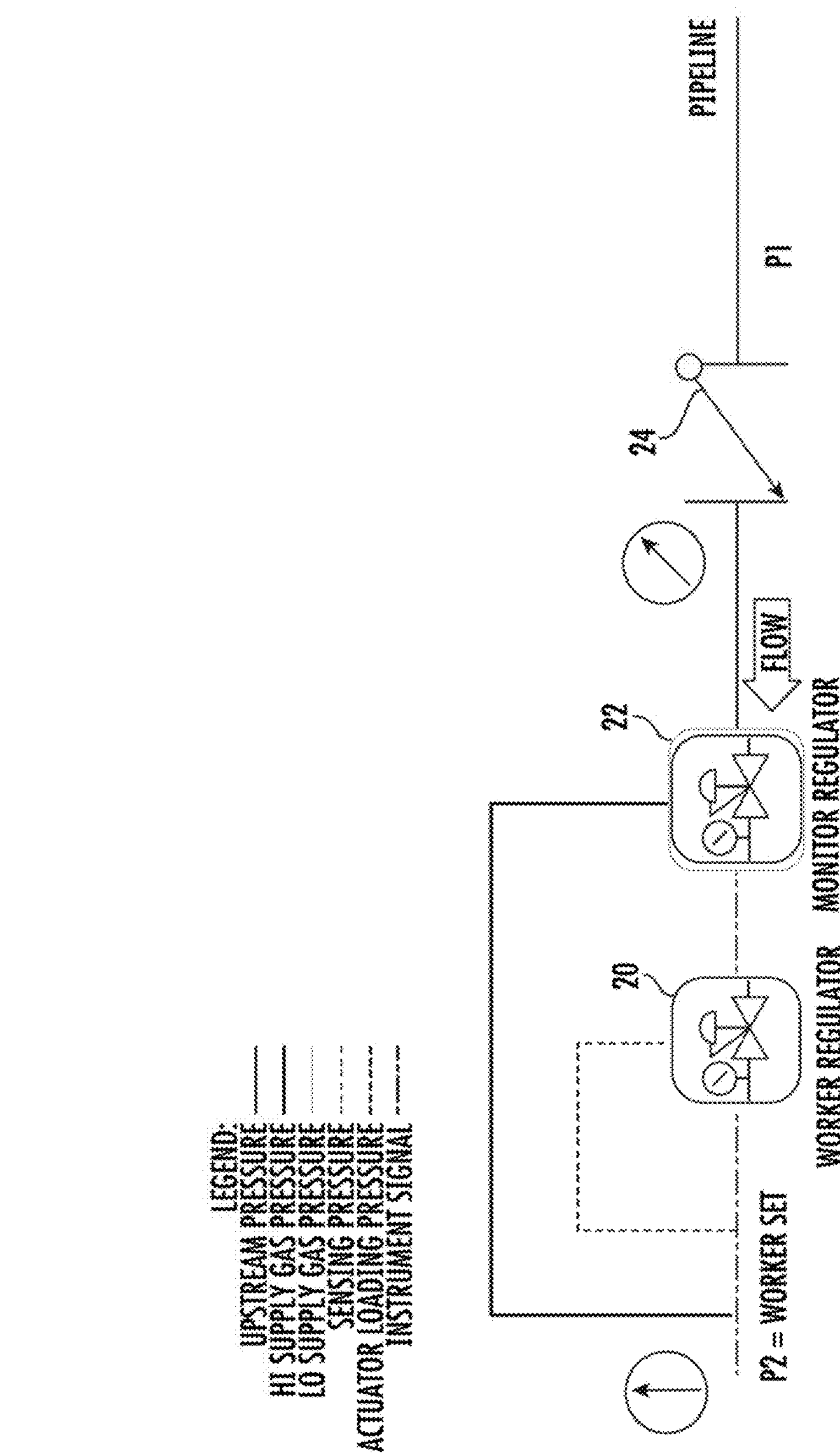


FIG. 6
PRIOR ART

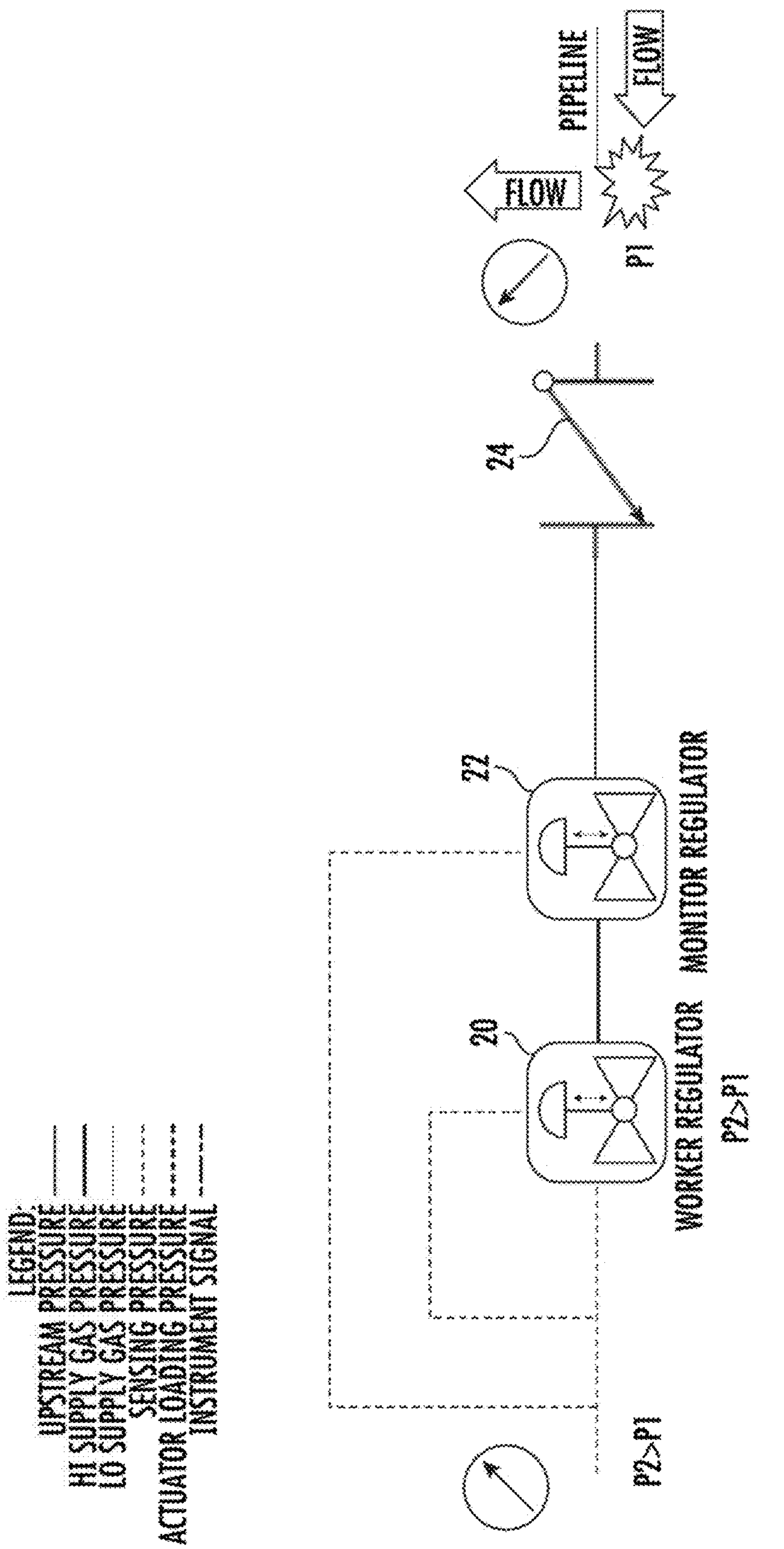


FIG. 7
PRIOR ART

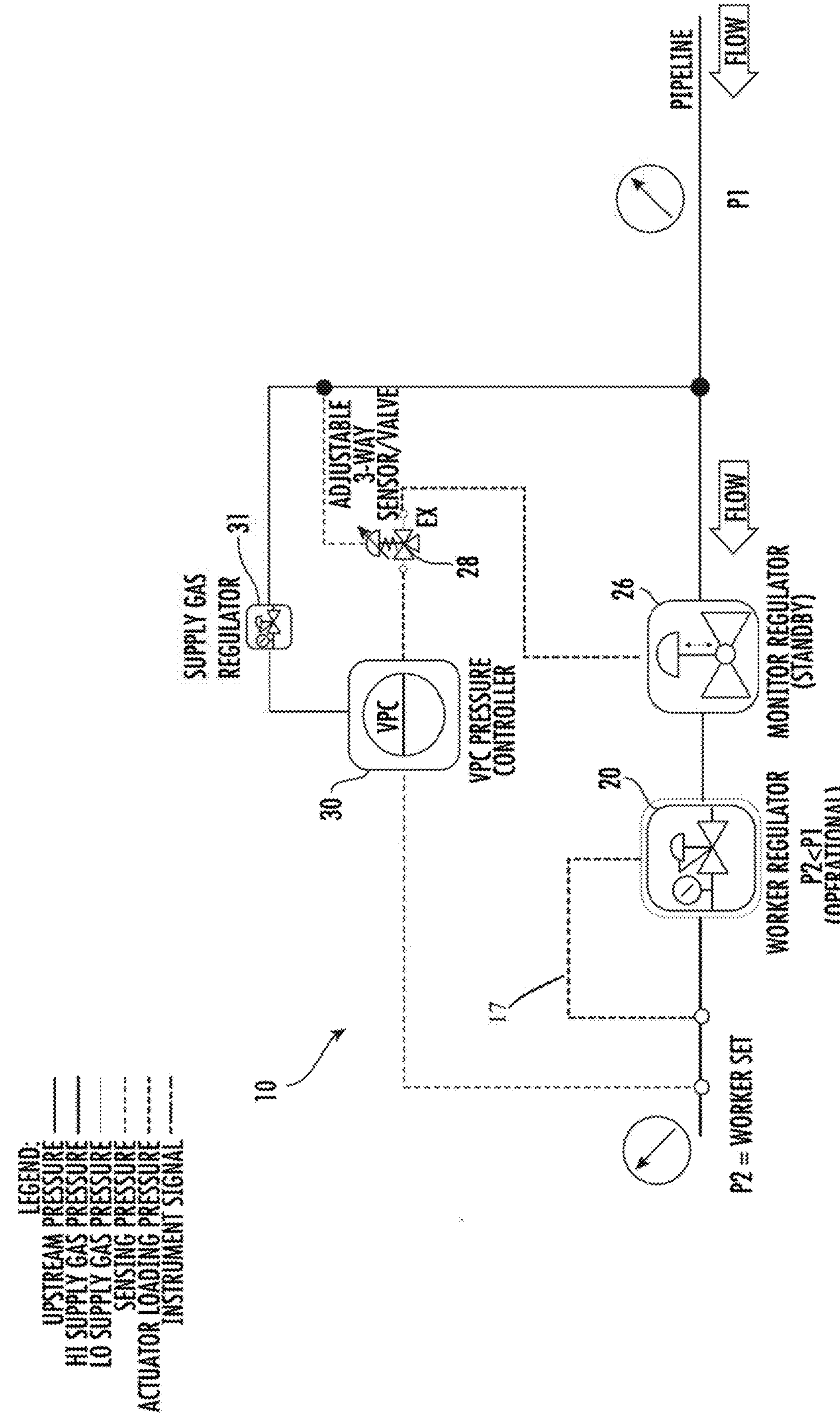

LEGEND:
UPSTREAM PRESSURE
HI SUPPLY GAS PRESSURE
LO SUPPLY GAS PRESSURE
SENSING PRESSURE
ACTUATOR LOADING PRESSURE
INSTRUMENT SIGNAL
10
17
P2 = WORKER SET
30
VPC
VPC PRESSURE CONTROLLER
SUPPLY GAS REGULATOR
31
ADJUSTABLE 3-WAY SENSOR/VALVE
EX
28
20
WORKER REGULATOR
P2<P1
(OPERATIONAL)
26
MONITOR REGULATOR
(STANDBY)
FLOW
P1
PIPELINE
FLOW

FIG. 8

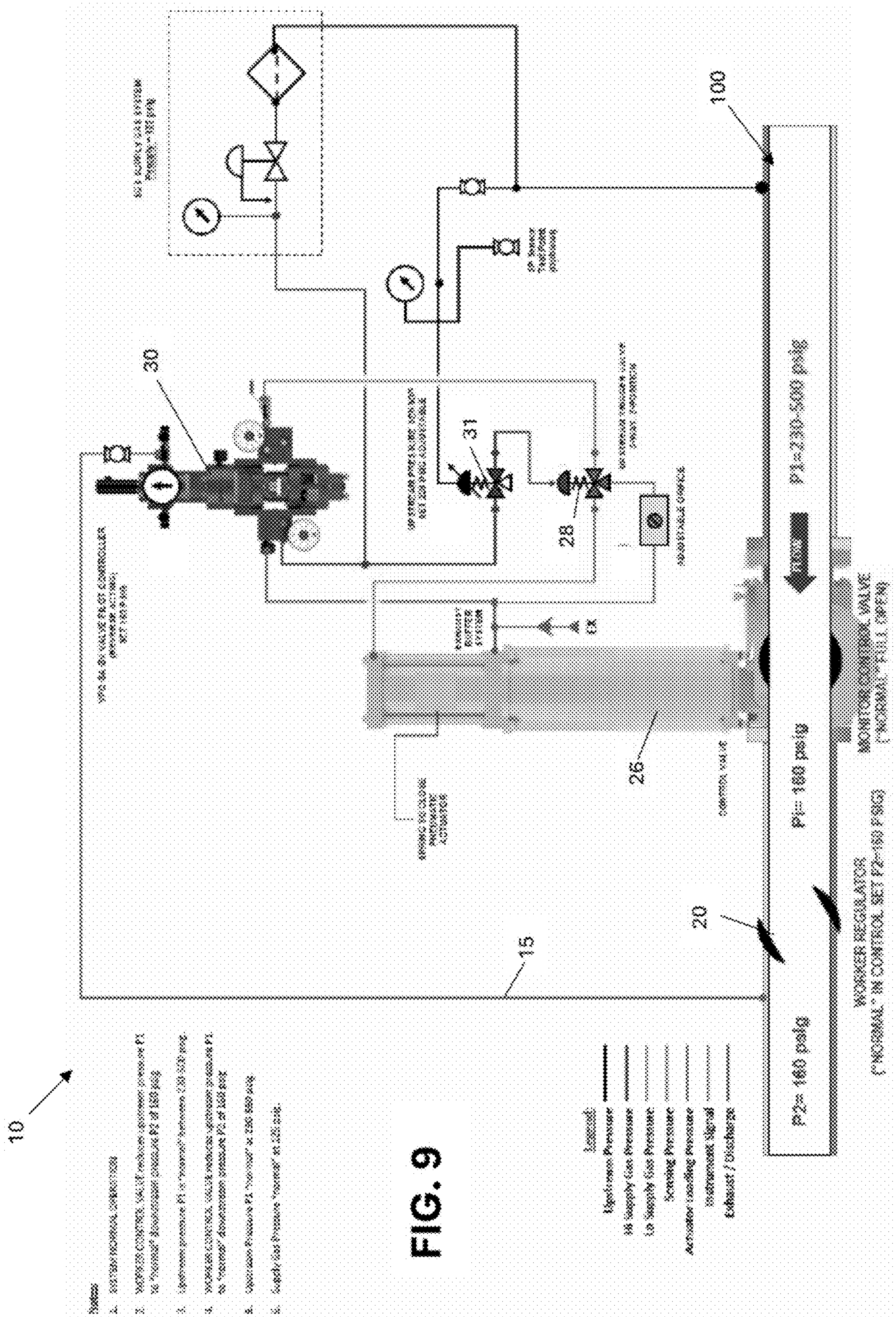

10
30
31
28
26
20
15
100
P1=230-500 psig
Pi= 160 psig
P2= 160 psig
WORKER REGULATOR
("NORMAL" IN CONTROL SET P2=160 PSIG)
MONITOR CONTROL VALVE
("NORMAL" FULL OPEN)
Legend:
Upstream Pressure
Hi Supply Gas Pressure
Lo Supply Gas Pressure
Sensing Pressure
Actuator Loading Pressure
Instrument Signal
Exhaust / Discharge

FIG. 9

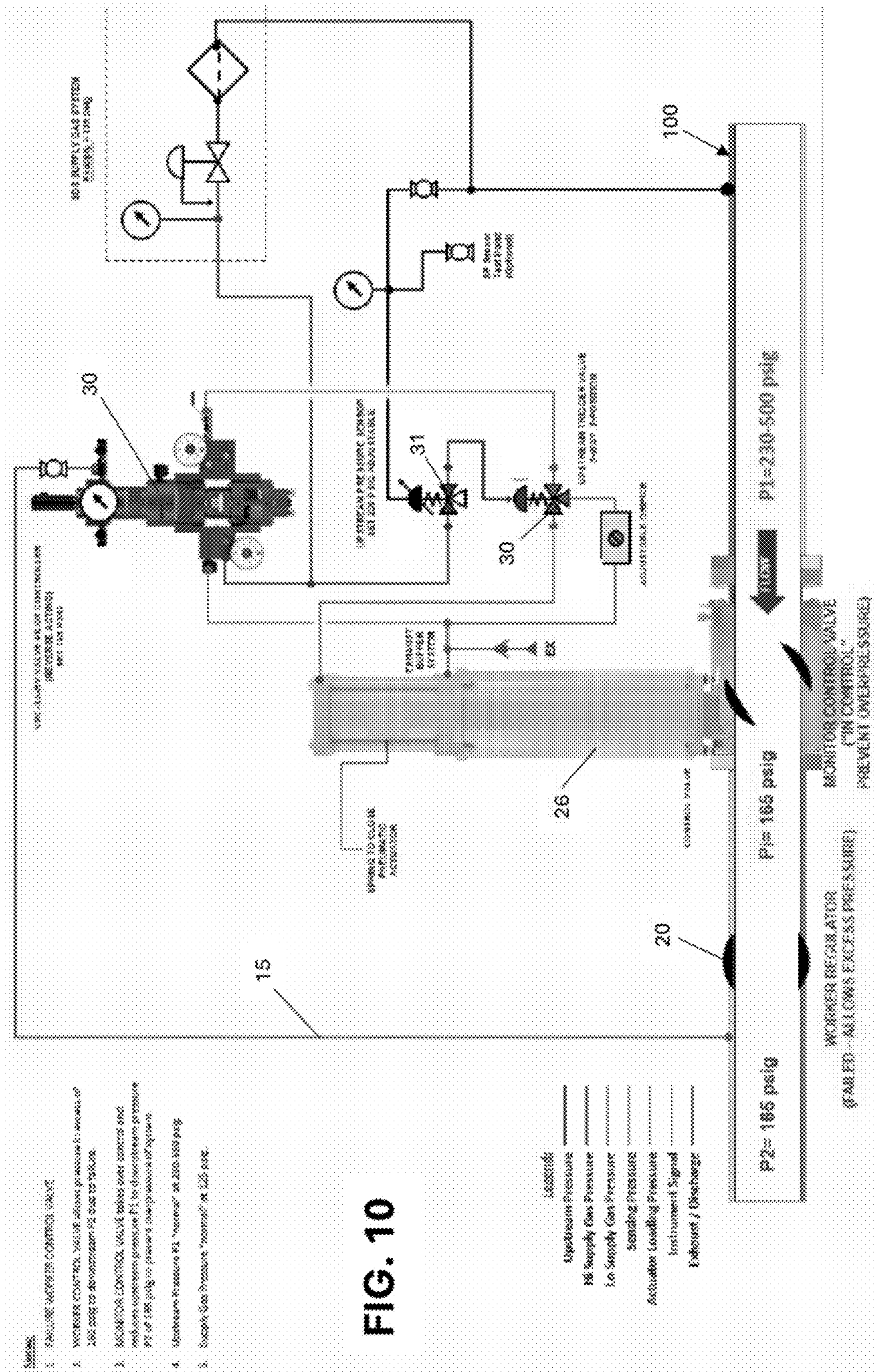

30
31
30
26
20
15
100
P1=230-500 psig
P/= 165 psig
P2= 165 psig
MONITOR CONTROL VALVE
("IN CONTROL"
PREVENT OVERPRESSURE)
WORKER REGULATOR
(FAILED - ALLOWS EXCESS PRESSURE)

FIG. 10

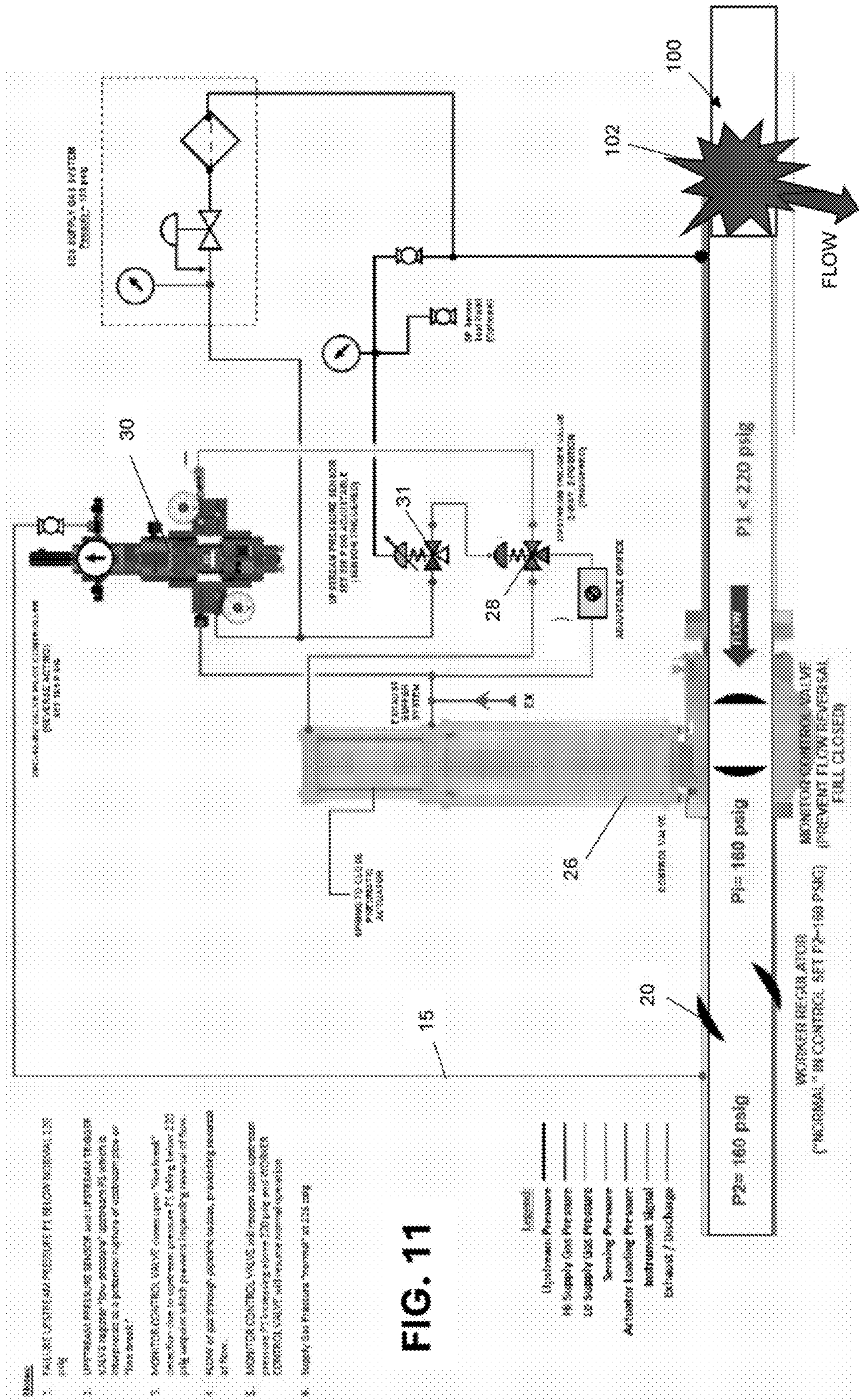


FIG. 11

METHOD OF BACKFLOW PREVENTION IN A GAS DELIVERY LINE

RELATED APPLICATION

[0001] The present application is a continuation of U.S. patent application Ser. No. 18/416,609, now U.S. Patent No. ______, titled "Monitor Control Valve With Backflow Prevention" filed on Jan. 19, 2024, which was a continuation-in-part of U.S. patent application Ser. No. 17/453,731 titled "Monitor Control Valve With Backflow Prevention" filed on Nov. 5, 2021, which claims the filing priority of U.S. Provisional Application No. 63/110,522, titled "Monitor Control Valve with Backflow Prevention" and filed on Nov. 6, 2020. The '609 application, the '731 application, and the '522 application are all hereby incorporated by reference.

TECHNICAL FIELD OF THE INVENTION

[0002] The present invention relates to control systems. More specifically, the invention relates to flow and pressure control systems for natural gas lines.

BACKGROUND OF THE INVENTION

[0003] In current natural gas line control systems, a significant drop in upstream or supply side pressure (P1) creates a potential backflow from the downstream or delivery side natural gas lines. This is an extremely undesirable occurrence. To prevent such a backflow condition, a backflow prevention valve—i.e., a check valve—is usually installed into the gas line upstream of the control valve (See FIG. **1**). This is a costly undertaking for natural gas suppliers and requires annual maintenance and certification of operation.

[0004] A significant upstream pressure drop (abnormal operating condition) is typically the result of a pipeline rupture or other significant event causing an upstream loss of gas. Normal upstream pressure is between 230 to 500 psig, while downstream pressure is generally maintained by regulators at 160 psig.

[0005] The use of a Monitor Control Valve has been used by the Assignee of this invention, VRG Controls, LLC., for many years as a way of providing "over-pressure" protection in a natural gas pipeline. FIG. **2** illustrates a normal gas flow where P2 is less than P1 due to pressure reduction maintained by a worker regulator **20**. As illustrated in FIG. **2**, a monitor regulator **22** positioned upstream of the worker regulator **20** is in standby mode during normal operation. As shown in FIG. **3**, the monitor regulator **22** takes over to throttle gas flow when the downstream pressure at P2 begins to exceed 160 psig. The onset of this over-pressure condition is often due to a failure of the worker regulator **20**. In the event of an upstream rupture, as illustrated in FIG. **4**, pressure at P1 drops significantly due to the loss of gas. When the pressure at P1 falls below pressure at P2, a reverse flow scenario is created. The worker regulator **20** and monitor regulator **22** cannot act to prevent the reverse flow, leading to a significant loss of product, loss of revenue, and unsafe conditions due to release of flammable media to atmosphere.

[0006] Unfortunately, the potential occurrence of flow reversal in a natural gas line has only been addressed by the addition of a backflow prevention valve—i.e., a check valve. The gas line arrangement illustrated in FIGS. **5-7** shows a typical check valve **24** positioned upstream of the two regulators, **20** and **22**. FIG. **5** shows normal gas line conditions, where P2 is less than P1 and gas flow is regulated by the worker regulator **20**. FIG. **6** illustrates a worker regulator failure, causing the monitor regulator **22** to take control of pressure regulation, much like the system of FIG. **3** explained above. In neither of these two scenarios does the check valve **24** interrupt flow. However, FIG. **7** illustrates a pipeline rupture upstream causing a drop of pressure at P1. The loss of pressure at P1 is almost immediate, but the check valve **24** will only close when the P1 pressure falls to or below P2 pressure. This equalization can take time after the occurrence of a line rupture.

[0007] Accordingly, a failsafe system is needed which can address both an over-pressure scenario as well as a reverse flow scenario in a gas line. Further, a failsafe system possesses an adjustable setpoint capable of responding to an upstream pressure drop well before pressure equalization is also desirable. This feature lessens the volume of gas lost to atmosphere in the event of a flow reversal due to rupture of the pipeline which benefits safety and lessens the potential of undersupplying gas consumers downstream of the pipeline rupture. Finally, a failsafe system which, unlike a check valve, is not costly to install and costly to maintain is most desirable.

[0008] Until the invention of the present application, these and other problems in the prior art went either unnoticed or unsolved by those skilled in the art. The present invention provides a system control for gas lines which performs multiple functions with associated devices potentially without requiring expensive retrofitting of valves in existing gas lines.

SUMMARY OF THE INVENTION

[0009] There is disclosed herein an improved natural gas line control system and methods which avoid the disadvantages of prior devices, systems and methods while affording additional structural and operating advantages.

[0010] Currently, a monitor control valve provides overpressure protection on pipeline. This means that if the "worker" (primary) control valve fails, then the monitor control valve takes control limiting downstream pressure. However, the disclosed monitor control valve system is altered to now include a "flow reversal prevention" feature which prevents reversal of flow if upstream pressure drops to predetermined setpoint. Addition of an upstream pressure sensor and trigger valve to prevent "flow reversal" is a new combination. The configuration eliminates the need for a customer to install a separate check valve component in the pipeline to prevent "flow reversal." The system may incorporate almost any type of control valve that can exhibit flow shutoff and has a mechanically actuated system that will guarantee closure of the valve upon loss of pneumatic supply pressure.

[0011] Generally speaking, the gas supply line control system comprises a control valve having an inlet, an outlet, and a mechanism for moving between an open and closed position to control gas flow from the inlet to the outlet, wherein gas flows from upstream to the inlet, to downstream through the outlet, a pressure sensor for determining an upstream side line pressure, and a trigger valve responsive to the pressure sensor for operating the control valve. The trigger valve closes the control valve mechanism when the pressure sensor determines an upstream pressure below a predetermined value to prevent reverse flow of gas.

[0012] The gas supply line control system work with one of almost any type of control valve that can exhibit flow shutoff and has a mechanically actuated system that will guarantee closure of the valve upon loss of pneumatic supply pressure.

[0013] A method for preventing flow reversal due to an upstream pressure drop in a natural gas supply line, the method comprising setting a threshold low pressure for the upstream gas line, continually sensing the upstream pressure, activating a trigger valve when the upstream pressure falls below the threshold low pressure, and closing a control valve in response to the trigger valve to prevent reversal of flow in the gas supply line.

[0014] These and other aspects of the invention may be understood more readily from the following description and the appended drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

[0015] For the purpose of facilitating an understanding of the subject matter sought to be protected, there are illustrated in the accompanying drawings, embodiments thereof, from an inspection of which, when considered in connection with the following description, the subject matter sought to be protected, its construction and operation, and many of its advantages should be readily understood and appreciated.

[0016] FIG. **1** is a schematic showing a prior art approach which splices a check valve into a gas line upstream of flow control regulators to prevent reverse flow in the event of a line rupture;

[0017] FIG. **2** is a schematic showing normal operation of a standard prior art gas line regulation system with a worker regulator and an upstream monitor regulator-worker control valve in control;

[0018] FIG. **3** is a schematic showing operation of a standard prior art gas line regulation system with a worker regulator and an upstream monitor regulator after failure of a worker regulator and activation of the monitor regulator;

[0019] FIG. **4** is a schematic showing operation of a standard prior art gas line regulation system with a worker regulator and an upstream monitor regulator after a gas line rupture upstream;

[0020] FIG. **5** is a schematic showing normal operation of a standard prior art gas line regulation system with a worker regulator and an upstream monitor regulator having an upstream check valve-worker valve in control;

[0021] FIG. **6** is a schematic of the system of FIG. **5** showing operation after failure of the worker regulator-monitor valve in control;

[0022] FIG. **7** is a schematic of the system of FIG. **5** showing operation after a pipeline rupture upstream-check valve engaged to prevent flow reversal;

[0023] FIG. **8** is an operational schematic showing normal operation of an embodiment of the disclosed gas line regulation system-worker valve in control;

[0024] FIG. **9** illustrates an embodiment of the disclosed control system during normal operation with a worker regulator controlling flow and a monitor control valve in a full-open position;

[0025] FIG. **10** illustrates the system of FIG. **9** showing operational changes after failure of the worker regulator with overpressure prevention logic of the monitor control valve throttling flow; and

[0026] FIG. **11** illustrates the system of FIG. **9** showing operational changes after a pipeline rupture upstream with backflow prevention logic of the monitor control valve in a full-closed position.

DETAILED DESCRIPTION OF THE INVENTION

[0027] While this invention is susceptible of embodiments in many different forms, there is shown in the drawings and will herein be described in detail at least one preferred embodiment of the invention with the understanding that the present disclosure is to be considered as an exemplification of the principles of the invention and is not intended to limit the broad aspect of the invention to any of the specific embodiments illustrated.

[0028] A schematic of the prior art use of a check valve **24** to prevent flow reversal in a gas line **100** is illustrated in FIGS. **1** and **5-7**. The addition of the check valve **24** is a time-consuming and costly undertaking requiring annual maintenance. The issue of flow reversal is more readily addressed by the present invention, which adds upstream pressure sensing and valve control to an existing control valve system.

[0029] Referring to FIG. **8**, operational parameters of the disclosed system are shown. The schematic of FIG. **8** is simplified for a general understanding of operation, as some actual components may be combined. The illustrated embodiment is directed to a control system **10** for a natural gas supply line **100**. The particular illustrated supply line **100** has a preferred upstream pressure (P1) in the range of 230 to 500 psig, while the downstream preferred pressure (P2) is 160 psig. Obviously, other set points for P1 and P2 pressures would be similarly addressed by the disclosed system **10**. Generally speaking, the control system **10** comprises a worker regulator valve **20**, a monitor control valve **26**, and a pressure controller (or valve pilot controller (VPC)) **30** with pressure sensing capabilities both upstream (P1) and downstream (P2). The monitor control valve **26** has an inlet side (upstream), an outlet side (downstream), and is controlled by trigger valve **28** via the pressure controller **30**. In the illustrated embodiment of FIGS. **9-11**, the trigger valve **28** is also coupled to an upstream pressure sensor **31**. Operationally, the monitor control valve **26** has a "full-open" default position, which can be (1) unchanged by the trigger valve **28**, (2) throttled via loading pressure from the trigger valve **28** in response to the controller **30** sensing increased P2, and (3) closed via loading pressure from the trigger valve **28** in response to pressure sensor **31** sensing decreasing P1 (below minimum setpoint).

[0030] For example, with respect to FIGS. **9-11**, the control system **10** maintains the set pressures, P1 and P2, by continuously monitoring both pressures. The upstream (supply side) pressure setpoints (e.g., 230 min. and 500 max.) are set at sensor **31**, which continually reads pressure on the upstream supply side of the monitor control valve **26**. Pressure controller **30** monitors P2 via line **15**, as shown-worker valve/regulator **20** also receives pressure feedback via line **17** of FIG. **8**. As P2 drops below a minimum setpoint, worker valve/regulator **20** is actuated to open to allow P2 to increase. Conversely, as P2 rises, worker valve/regulator **20** throttles flow to bring P2 back to 160 psig. When P1 is within the setpoint limits and P2 is at 160 psig, normal operation of the worker valve **20** occurs and the monitor valve **26** is at full-open position.

[0031] As can be seen in FIG. **9**, the system **10** is in normal operation. The worker regulator **20** is used to reduce delivery side pressure of the preferred "normal" upstream pressure (P1) from 230-500 psig to a preferred "normal" downstream delivery pressure (P2) of 160 psig. The monitor control valve **26** is in a full-open position (default), allowing the worker valve **20** to control flow pressure.

[0032] Referring now to FIG. **10**, the schematic illustrates control failure of the worker valve/regulator **20**. The failure of the worker valve **20** allows excess pressure in the downstream flow (i.e., delivery side) to reach a level greater than the preferred "normal" 160 psig. If allowed to continue, this scenario with reach what is called an "over-pressure" condition. This can be dangerous to residences and businesses which may not be equipped to regulate such high gas pressures. To prevent further climbing of the downstream (P2) pressure, the VPC pressure controller **30** senses the downstream pressure increase and transmits this pressure to the monitor trigger valve **28**. The upstream pressure sensor **31** also transmits the upstream pressure P1 to the 3-way trigger valve **28**, which then acts on the monitor valve **26** as a result of the climbing P2, to throttle flow. This continues to bring P2 back to the targeted 160 psig.

[0033] A final scenario is illustrated by the embodiment of FIG. **11**. Here the schematic presents a situation where an upstream event (e.g., a line rupture) **102** has caused a drop in upstream pressure (P1<220 psig). While the worker regulator **20** is operating normal (i.e., throttling gas flow) to maintain downstream pressure (P2) at the preferred 160 psig, the low and dropping upstream pressure will eventually cause a flow reversal in the gas line—i.e., backflow—when P1<P2 is achieved. The upstream pressure sensor **31**, which is set to 220 psig—i.e., below the lowest preferred "normal" upstream pressure of 230 psig—is triggered and signals the pressure controller **30** and/or the trigger valve **28** which then sends a loading pressure to the monitor valve **26**. The monitor control valve **26** is actuated to a full-closed position, thereby preventing reverse flow in the gas line **100**.

[0034] The control valve **26** is able to close much earlier than a check valve of the prior art (see FIG. **1**), thereby preventing a greater loss of product. The control valve **26** may be a pressure regulator but must have a guaranteed physical close to be used in the disclosed system **10**.

[0035] The monitor control valve **26** used for the disclosed system **10** may be either a rotary control valve or a linear control valve as manufactured and sold by Assignee, VRG Controls, LLC. (see https://www.vrgcontrols.com/control-valves). Further, while all the embodiments illustrated are directed to a natural gas supply line, it should be understood that the principles of the invention can be more broadly applied to most any fluid delivery system where reverse flow presents an issue.

[0036] The matter set forth in the foregoing description and accompanying drawings is offered by way of illustration only and not as a limitation. While particular embodiments have been shown and described, it will be apparent to those skilled in the art that changes and modifications may be made without departing from the broader aspects of applicants' contribution. The actual scope of the protection sought is intended to be defined in the following claims when viewed in their proper perspective based on the prior art.

What is claimed is:

1. A method for preventing flow reversal in a natural gas supply line due to an upstream pressure drop, the method comprising:

throttling a gas pressure through the natural gas supply line, using a worker regulator positioned downstream of a monitor control valve, to maintain a downstream delivery side gas flow pressure;

setting a threshold low pressure (X) for gas flow upstream of the monitor control valve;

continually sensing an actual gas flow pressure, P_1, at a point upstream of the monitor control valve;

continually sensing an actual gas flow pressure, P_2, at a point downstream of the monitor control valve, wherein the threshold low pressure (X) is set to be greater than the actual gas flow pressure (P_2); and

closing the monitor control valve when the actual flow pressure upstream of the monitor control valve, P_1, falls below the threshold low pressure (X) to prevent reversal of flow in the gas supply line.

2. The method of claim **1**, further comprising re-opening the closed monitor control valve when the actual flow pressure upstream of the monitor control valve, P_1, exceeds the threshold low pressure (X).

3. The method of claim **1**, wherein the monitor control valve is a rotary control valve.

4. The method of claim **1**, wherein the monitor control valve is a linear control valve.

5. The method of claim **1**, further comprising throttling gas flow with the monitor control valve when actual gas flow pressure at a point downstream of the monitor control valve, P_2, exceeds a predetermined pressure value.

6. The method of claim **5**, wherein the threshold low pressure (X) is greater than 220 psig.

7. A gas supply line control system comprising:

a worker regulator for maintaining a downstream pressure in a gas supply line at a target pressure;

a monitor control valve upstream of the worker regulator and having an inlet, an outlet, and a mechanism for moving within a range between and including a fully open position and a fully closed position to control gas flow in the gas supply line, wherein gas flows from upstream to the inlet, to downstream through the outlet;

a first pressure sensor for determining a flow pressure upstream of the monitor control valve inlet; and

a trigger valve responsive to the sensor;

wherein the trigger valve triggers the monitor control valve mechanism to fully close to prevent reverse gas flow when the upstream flow pressure falls below a first predetermined value.

8. The gas supply line control system of claim **7**, further comprising a second pressure sensor for determining a flow pressure downstream of the monitor control valve outlet.

9. The gas supply line control system of claim **8**, wherein the monitor control valve throttles the gas flow when the second pressure sensor detects a downstream pressure exceeding a second predetermined value.

10. The gas supply line control system of claim **9**, wherein the second predetermined value is 160 psig.

* * * * *